ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A Hospital Management System allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

Tasks:
a) Identify all actors involved in the system.
b) List the primary use cases.
c) Define relationships (include, extend, generalization).
d) Construct a complete Use Case Diagram representing system functionality.

a) Actors Involved

1. **Patient** – Books appointments, views records
2. **Doctor** – Updates medical records, generates prescriptions
3. **Administrator** – Manages users and resources
4. **Receptionist** – Registers patients, schedules appointments
5. **Pharmacist** – Manages medicines and prescriptions

b) Primary Use Cases

1. Register Patient
2. Book Appointment
3. View Medical Records
4. Update Medical Records
5. Generate Prescription
6. Manage Hospital Resources
7. Manage Medicines

c) Relationships

Include (<<include>>):

- Book Appointment → Patient Verification
- Generate Prescription → View Medical Records

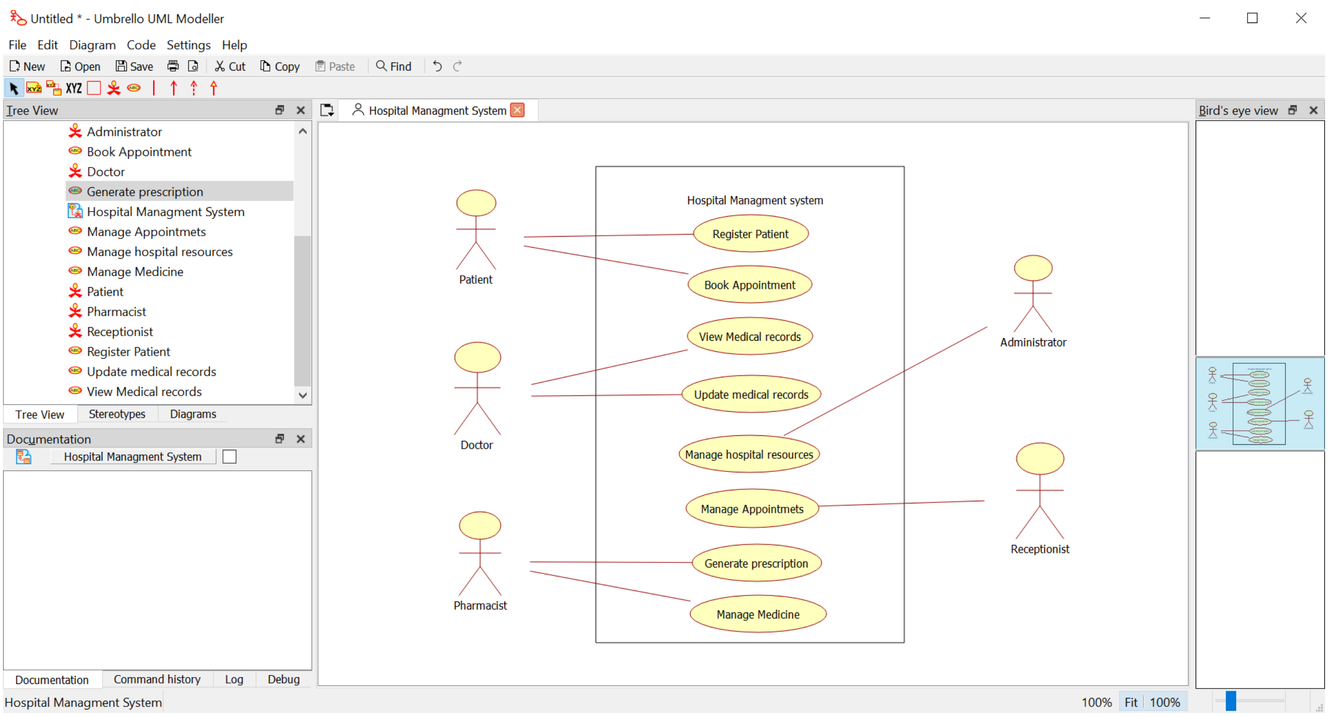
Extend (<<extend>>):

- Cancel Appointment → Book Appointment

Generalization:

- User → Patient, Doctor, Administrator, Receptionist, Pharmacist

d)



2). Class Diagram Problem

A Library Management System contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

Tasks:

a) Identify classes with attributes and methods.

b) Define relationships (association, aggregation, inheritance).

c) Specify multiplicity between classes.

d) Design a detailed Class Diagram for the system.

a) Classes, Attributes & Methods

Book

Attributes: Book_ID, Title, Author, Status

Methods: SearchBook(), UpdateStatus()

Member

Attributes: Member_ID, Name, Email

Methods: BorrowBook(), ReturnBook()

Librarian

Attributes: Librarian_ID, Name, Contact

Methods: AddBook(), RemoveBook()

IssueRecord

Attributes: Issue_ID, Issue_Date, Return_Date

Methods: IssueBook(), ReturnBook()

b) Relationships

- **Association:** Member borrows Book through IssueRecord.
- **Aggregation:** Library contains Books and Members.
- **Inheritance:** User → Member, Librarian

User

|

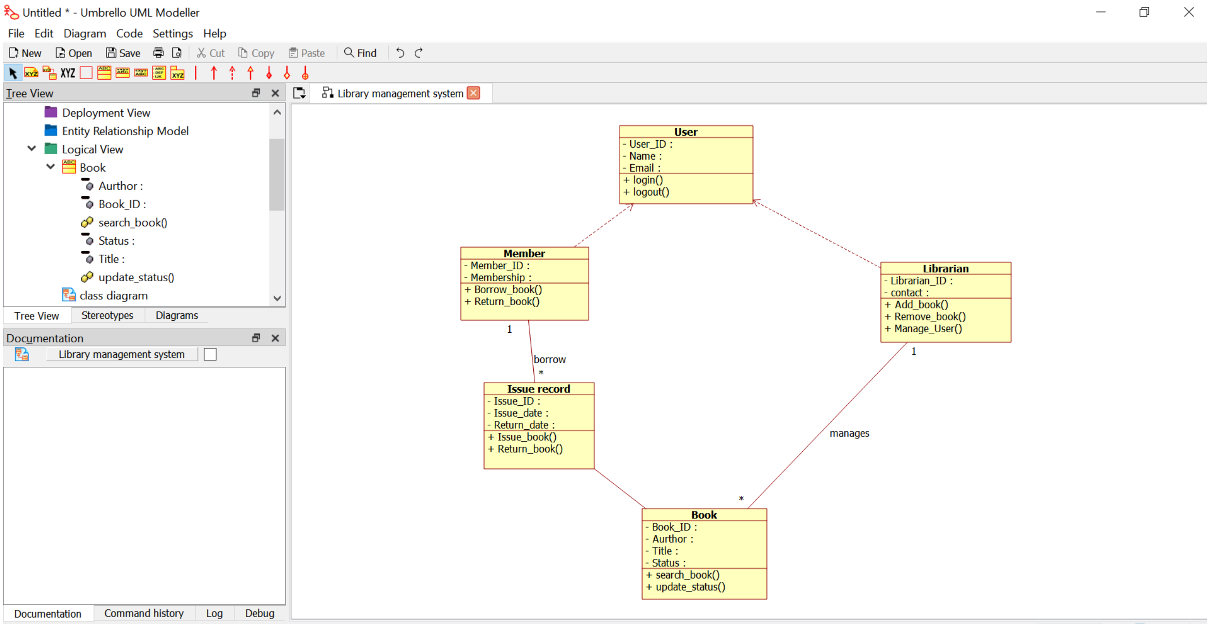
| |

Member   Librarian

c) Multiplicity

- Member **1 : *** IssueRecord
- Book **1 : *** IssueRecord
- Librarian **1 : *** Book

d)



7. Deployment Diagram Problem

A **Smart Traffic Monitoring System** uses roadside sensors to collect vehicle data. The data is transmitted to a central cloud server where traffic authorities monitor congestion through a web dashboard. Alerts are generated during traffic violations and accidents.

Tasks:

a) Identify all hardware nodes involved in the system.

b) Specify software components deployed on each node.

c) Define communication links between nodes.

d) Construct a complete Deployment Diagram representing the system architecture.

a) Identify all hardware nodes involved in the system

The hardware nodes are:

1. Roadside Sensor
2. IoT Gateway
3. Cloud Server
4. Traffic Control PC
5. Web Browser (User Device)

b) Specify software components deployed on each node

Hardware Node

- Roadside Sensor
- IoT Gateway
- Cloud Server
- Traffic Control PC
- Web Browser

Software Component

- Vehicle Detection Software
- Data Collection Service
- Traffic Monitoring System, Database, Alert Service

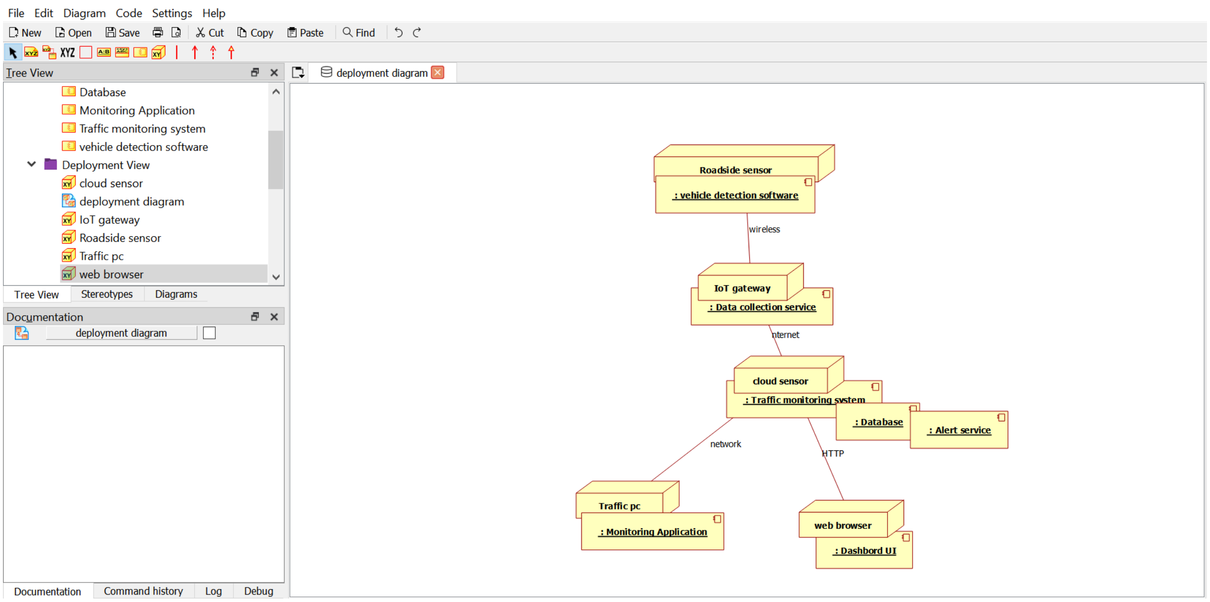
Software Component

- Monitoring Application
- Dashboard User Interface (UI)

c) Define communication links between nodes

- **Roadside Sensor ↔ IoT Gateway** – Wireless Communication
- **IoT Gateway ↔ Cloud Server** – Internet
- **Cloud Server ↔ Traffic Control PC** – Network (LAN/WAN)
- **Cloud Server ↔ Web Browser** – HTTPS

d)



6. Component Diagram Problem

An **Online Learning Platform** consists of modules such as User Management, Course Management, Assessment Module, Content Delivery Module, and Notification Service.

Tasks:
a) Identify major system components.
b) Define interfaces between components.
c) Show dependencies among modules.
d) Construct a Component Diagram representing system architecture.

a) Major System Components

1. User Management Module – Handles user registration, login, and profiles.
2. Course Management Module – Manages courses, enrollment, and course details.
3. Content Delivery Module – Provides learning materials like videos and documents.
4. Assessment Module – Manages quizzes, exams, and results.
5. Notification Service – Sends alerts, reminders, and updates.
6. Database – Stores user, course, content, and assessment data.

b) Interfaces Between Components

- User Management ↔ Course Management: User Authentication Interface
- Course Management ↔ Content Delivery: Content Access Interface
- Course Management ↔ Assessment Module: Assessment Interface
- Assessment Module ↔ Notification Service: Result Notification Interface
- All Modules ↔ Database: Data Access Interface

c) Dependencies Among Modules

- User Management depends on Database for storing user details.
- Course Management depends on User Management for user access.
- Course Management depends on Content Delivery for providing materials.
- Assessment Module depends on Course Management for course assessments.
- Assessment Module depends on Notification Service for sending results.
- All modules depend on Database for data storage.

d)